



Developmental biology of medaka fish (*Oryzias latipes*) exposed to alkalinity stress

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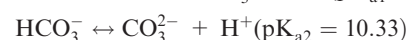
Summary

Alkalinity stress is common in cultured aquatic animals and considered to be one of the major stress factors for fishes when they are transferred to saline-alkali waters. To evaluate potential effects of alkalinity on the developmental biology of *Oryzias latipes*, fertilized eggs, larvae and breeding fish were exposed to different carbonate alkalinity concentrations of 1.5–64.5 meq l⁻¹, for 9, 120, and 60 days, respectively. The mortality of embryos significantly increased when exposed to the high concentrations (16.5–64.5 meq l⁻¹). Although more than 50% of survived embryos hatched in 16.5 and 31.4 meq l⁻¹ concentrations of carbonate alkalinity, most were not able to swim up after hatching. Morphological abnormalities such as coagulated embryos, halted embryo development, and hatching failure were observed at stages 15, 29–33 and 38 in high concentrations (31.4, 64.5 meq l⁻¹). Almost all larvae in 16.5 and 31.4 meq l⁻¹ treatments died 70 d post-hatch. Growth of juveniles exposed to carbonate alkalinity of 5.3 and 8.8 meq l⁻¹ was not significantly different at 70 d and 120 d post-hatch. The number of eggs released by breeders, the fertilization rate and the hatching rate of eggs were significantly lower in the 31.4 meq l⁻¹ treatment than in other treatments. Although medaka are capable of surviving in high alkalinities (31.4, 64.5 meq l⁻¹) for an extended period of time, these conditions are stressful to the fish, especially at the embryonic and reproductive stages.

Introduction

Saline-alkali land area covers 0.95 billion hectares throughout the world, accounting for 1/3 of the total land area (Sumner and Naidu, 1998). Saline-alkali land covers about 99.13 million hectares of China, distributed in the northwest, northeast and north as well as in areas along the north bank of the Yangtze River, including the Xinjiang, Mongolia, and Ningxia autonomous regions; Qinghai, Gansu, Shanxi, Hebei, Shandong, Henan, Jiangsu, Liaoning, Jilin, Heilongjiang provinces; and the cities of Beijing and Tianjin, etc. (Wang, 1993). About 45.87 million hectares of saline-alkali water are distributed throughout these areas, most of which is athalassic saline water characterized by a high pH value of about 8.8, high carbonate alkalinity (CA) concentrations, and various types of ion imbalances. These stressful environmental conditions allow few organisms to survive. Hence, most of these saline-alkali water areas are desolate (Wang et al., 2003). High CA is one of the major stress factors for living organisms in the saline-alkali water areas. Two carbonates, NaHCO₃ and Na₂CO₃, are

responsible for the high carbonate alkalinities, and decompose as:



According to the above equations, CO₂ is the predominant component at a pH lower than 6.35, while HCO₃⁻ is the predominant component at a pH of 6.35–10.33, and CO₃²⁻ is predominant at a pH higher than 10.33. When the pH is between 7 and 9, the bicarbonate ion is most important and responsible for buffering aquatic ecosystems. The equations also show that CA stresses to fish are related to acid-base regulation (Thomas and Poupin, 1985), breadth of carbon dioxide (Truchot and Forgue, 1998), as well as dependent on ionic regulation and osmoregulation in fish (Gossl et al., 1994).

Various researchers have studied high carbonate alkalinity effects of CA on fishes. However, those studies were focused on the acid-base regulation of fishes injected or perfused with bicarbonate (Perry and Gilmour, 2006). Thomas and Poupin (1985) found that high levels of CA induce a situation of mixed respiratory and metabolic alkalosis in adult rainbow trout. However, low CA offers the fish gill carbonate buffering conditions, necessary to catalyse the hydration of excreted CO₂ in the gill water and the location of carbonic anhydrase on the water side of the respiratory epithelium (Truchot and Forgue, 1998).

Medaka (*Oryzias latipes*) is a model vertebrate for the study of developmental and evolutionary biology. It has been widely used for studies on the effects of radiation (Neues et al., 2007), carcinogens (Liu et al., 2003) and chemicals (Scholz and Mayer, 2008). Medaka is an attractive model organism to evaluate the life-cycle toxicity in both parental and progeny generations because of its short maturation time (Wittbrodt et al., 2002), i.e. within 6–8 weeks after hatching. In addition, medaka can adapt to saltwater environments. Our recent study shows that adult *O. latipes* are able of surviving at 80 ppt salinity through gradual acclimation, and may offer a new model for studying mechanisms of seawater adaptation.

The stress brought about by CA has not been studied extensively and is seldom reported. The objectives of this study were to investigate the effects of carbonate alkalinity on embryonic and early larval development as well as the reproductive performance of *Oryzias latipes* in order to evaluate the eventual application of medaka as a model animal in studying CA tolerance.

Materials and methods

Test fish

Experimental medaka were maintained for two generations in our laboratory. About 200 pairs of breeders [910 ± 57 mg weight and 42 ± 0.7 mm total length (TL), $n = 30$] were kept for at least 2 weeks prior to the experiment in 50-L glass tanks with filtered tap water (pH: 7.45 ± 0.4 , $n = 30$; dissolved oxygen: 5.75 ± 0.8 mg l^{-1} , $n = 30$; ammonia: < 0.1 mg l^{-1} , $n = 30$; nitrite: 5 ± 1 mg l^{-1} , $n = 30$) at $26 \pm 1^\circ\text{C}$. The tanks were cleaned by siphon to remove debris, and 50% of the water was changed daily.

During the mating period the fishes were placed under a photoperiod of 16 : 8-h light : dark and fed pellets and nauplii of *Artemia* sp. twice per day. Eggs spawned from each female fish were collected and separated individually within a few hours after fertilization and placed in a petri dish containing about 50 ml of Ringer's solution with 1% methylene blue. Fertilization and development (Doncel et al., 2005) were regularly monitored every 24 h under a light stereomicroscope (SZX16; Olympus, Tokyo, Japan).

CA exposure design

The CA solution was prepared by adding NaHCO_3 and Na_2CO_3 to Ringer's solution or filtered tap water for the embryonic stage and the other stage test, respectively. The CA solution was maintained in plastic tanks for 24 h before use. The test tank for adult fish was designed to contain about 10 L of the test solution. Tanks ($14 \times 10 \times 8.5$ cm) containing 10 L

of the test solution were used for the reproduction test, each containing one male and one female. The 6-well multiplates (Sigma-Aldrich product #M856 Sigma-Aldrich Corp., St. Louis, MO, USA) containing about 15 ml of the test solution in each cell were used for the embryo test. The test solutions in the test plates and tanks were prepared daily and changed 100% daily. The fresh solution was prepared daily.

The treatment concentrations of CA at 0 to 64 meq l^{-1} (Table 1) used for this experiment were determined based on a 7–10-d preliminary exposure of the various developmental stages of medaka. Photoperiod was 16 h light : 8 h dark. Water temperature was controlled with a thermostat at 26°C . For the reproductive stage, the temperature was increased to 28°C to stimulate spawning. Test equipment and tanks were cleaned daily to prevent bacterial bloom and accumulation of CA on the walls of the test chambers. Residual food and feces in the test tanks were siphoned out daily.

Biology protocols

Embryonic stage Exposure was initiated at the 16-cell stage (stage 6, 3 h post-fertilization (Doncel et al., 2005)). Each treatment had four replicates, each with 15 embryos placed in individual test plates (Fig. 1, Phase 2). The embryos were observed daily under a stereomicroscope until all surviving embryos hatched; any dead embryos were discarded immediately.

Larval–juvenile stage The newly-hatched medaka larvae from the embryonic stage test were cultured in different carbonate

Table 1

Designed and measured concentrations (DC, MC) of carbonate alkalinity (CA) ($n = 30$)

DC (meq l^{-1})	0 (Control)	4	8	16	32	64
MC (meq l^{-1})	1.5 ± 0.1	5.3 ± 0.3	8.8 ± 0.4	16.5 ± 0.4	31.4 ± 0.5	64.5 ± 0.9

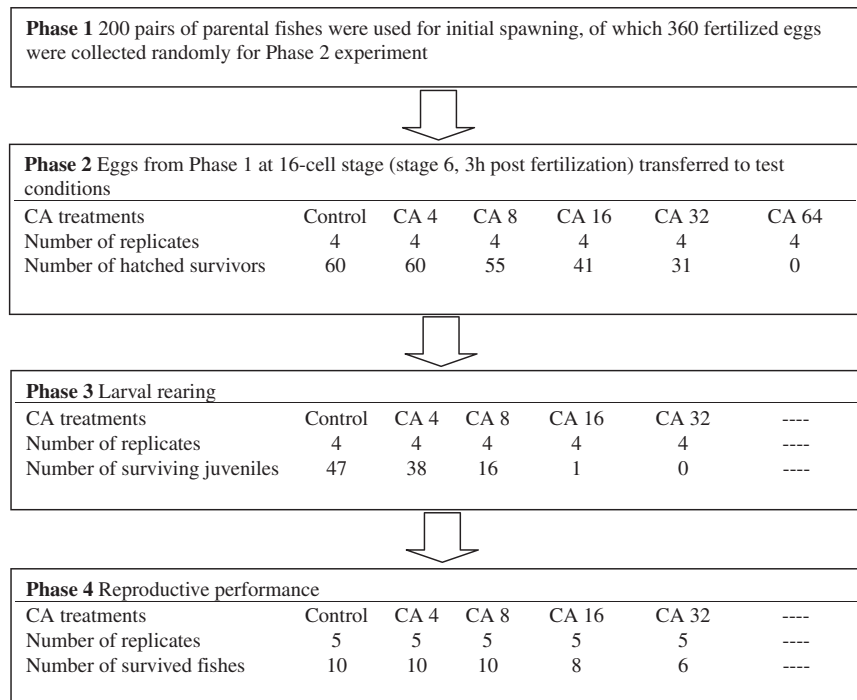


Fig. 1. Flow chart of experimental design. CA4, CA8, CA16, CA32, CA64 present carbonate alkalinity (CA) concentrations of: 4, 8, 16, 32, and 64 meq l^{-1} , respectively

alkalinity waters (Fig. 1, Phase 3). The larvae were fed with *Artemia* sp. nauplii (24 h post-hatch) at approximately 5–10 nauplii ml⁻¹ twice daily. Larval behavior and mortality were observed daily until 120 d post-hatch (21.9 ± 8.9 mm, n = 30). If any mortality occurred, dead larvae were removed upon daily inspection. Total length and body weight were measured and growth rate was calculated (Taylor et al., 2005):

Specific growth rate of total length (SGR_L, % d⁻¹)

$$= (\ln L_2 - \ln L_1) / (t_2 - t_1);$$

Specific growth rate of body weight (SGR_W, % d⁻¹)

$$= (\ln W_2 - \ln W_1) / (t_2 - t_1);$$

Where L₂ and L₁ were total length on t₂ and t₁; W₂ and W₁ were body weight on t₂ and t₁; t₂–t₁ was the interval (days) between two measurements.

Reproductive stage One-year-old medaka breeders were used to evaluate the reproductive performance (Fig. 1, Phase 4). Sexes were determined by their morphology, mainly the difference in the abdomen portion. Five mating pairs from each treatment and the controls were used for the experiment, each test tank containing one pair. The number of eggs spawned per pair, fertilization rate and hatching rate were examined. Water temperature was controlled at 28°C to stimulate spawning. Eggs spawned from each female were collected, counted, and assessed for viability microscopically. The test lasted 50 consecutive days; at the end of the test, body weight and total length of the fishes were measured, gonads removed and weighed, and the GSI was calculated as: GSI = 100 × gonad weight/bodyweight.

Determination of CA concentration

The concentration of CA was determined every 2 weeks throughout the exposure period with the acidimetric titration end-point (pH = 4.3) method, expressed as meq l⁻¹, 1 meq l⁻¹ = 50.25 mg CaCO₃ l⁻¹.

Statistical analysis

One-way analysis of variance (ANOVA) was applied to evaluate the differences in egg mortality, hatchability, swim-up failure, time to hatch, coagulated embryos, halted development, hatching failure, larval mortality, total length, body weight,

SGR_L, SGR_W, number of eggs spawned, and GSI. All variables were tested for homogeneity and normality using Levene's test and Shapiro–Wilk's test, respectively. When ANOVA effects were significant, the Student-Newman-Keuls (SNK) multiple comparison was used to compare the difference between the different means. All statistical analyses were conducted with SAS 8.0 (SAS Institute Inc, 1988). Differences were considered to be significant at the P = 0.05 level.

Results

Embryonic stage

Embryonic development and hatching of medaka eggs were greatly affected by carbonate alkalinity exposure. Egg mortality increased with increasing CA concentration (Table 2). There was 100% mortality at the highest concentration (64.5 meq l⁻¹), significantly different from the control (P < 0.05). At concentrations of 16.5 and 31.4 meq l⁻¹, mortality rates (29.2 and 48.3%, respectively) were also significantly higher than in the control (P < 0.05). Hatching rates in all treatments were > 50%, except in the 64.5 meq l⁻¹ treatment; however, < 50% of hatched larvae swam up successfully in the 16.5, 31.4 and 64.5 meq l⁻¹ treatments, significant lower than in the control (P < 0.05). Egg hatching times were similar in all treatments, varying from 7 to 9 days. Some lethal morphological abnormalities such as coagulated embryos, halted development, and hatching failure were found at stages 15, 29–33 and 38 (Table 3). About 16.8 and 22.5% of embryos were coagulated at stage 15 at concentrations of 31.4 and 64.5 meq l⁻¹, significantly higher than in the control (P < 0.05). During development from stage 29 to stage 33, halted development occurred at 16.5, 31.4 and 4.5 meq l⁻¹, with 15.5, 16.3, and 35.0% significantly higher than in the control (P < 0.05). Almost 50% of embryos failed to hatch at the final stage of embryogenesis in the 64.5 meq l⁻¹ treatment.

Larval–juvenile stage

Almost all fishes in the 16.5 and 31.4 meq l⁻¹ treatments died after 70 d post-hatch. In addition, the mortalities in 5.3 and

Table 2
Effect of carbonate alkalinity (CA) on embryogenesis and egg hatching of *Oryzias latipes*. Mean ± SE. (4 replicates, 15 eggs per replicate) of egg mortality, hatchability, swim-up failure, and hatch time of fertilized eggs

CA (meq l ⁻¹) ¹	Egg mortality (%)	Hatchability (%)	Swim-up failure (%)	Time to hatch (h)
1.5	0.0 ± 0.0 ^a	100.0 ± 0.0 ^d	0.0 ± 0.0 ^a	8.0 ± 0.4 ^a
5.3	0.0 ± 0.0 ^a	100.0 ± 0.0 ^d	0.0 ± 0.0 ^a	8.1 ± 0.1 ^a
8.8	0.0 ± 0.0 ^a	91.7 ± 8.3 ^d	50.0 ± 16.7 ^b	7.8 ± 0.1 ^a
16.5	29.2 ± 4.2 ^b	70.9 ± 4.2 ^c	62.5 ± 12.5 ^b	8.7 ± 1.4 ^a
31.4	48.3 ± 5.5 ^c	51.7 ± 5.5 ^b	62.5 ± 8.0 ^b	7.4 ± 0.4 ^a
64.5	100.0 ± 0.0 ^d	0.0 ± 0.0 ^a	–	–

¹Mean measured concentrations.

Different letters in each column = significant differences (P < 0.05) after multiple comparisons with Student-Newman-Keuls method.

Table 3
Morphological abnormalities in embryogenesis of *Oryzias latipes* exposed to carbonate alkalinity (CA) stress. Mean ± SE. (4 replicates, 15 eggs per replicate)

CA (meq l ⁻¹) ¹	S15 Coagulated embryo (%)	S29–S33 Halted embryo (%) development	S38 Hatching failure (%)
1.5	0.0 ± 0.0 ^a	0.0 ± 0.0 ^a	0.0 ± 0.0 ^a
5.3	0.0 ± 0.0 ^a	0.0 ± 0.0 ^a	0.0 ± 0.0 ^a
8.8	0.0 ± 0.0 ^a	0.0 ± 0.0 ^a	0.0 ± 0.0 ^a
16.5	0.0 ± 0.0 ^a	15.5 ± 5.4 ^b	13.3 ± 4.5 ^b
31.4	16.8 ± 5.9 ^b	16.3 ± 5.5 ^b	14.5 ± 5.2 ^b
64.5	22.5 ± 2.5 ^b	35.0 ± 2.9 ^c	42.5 ± 2.5 ^c

¹Mean measured concentrations.

Different superscript letters in each column = significant differences (P < 0.05) after multiple comparisons with Student-Newman-Keuls method.

S15 = stage 15 in the embryogenesis of *O. latipes* about 17 h after fertilization; S29 = stage 29 in the embryogenesis of *O. latipes* about 24 h after fertilization; S33 = stage 33 in the embryogenesis of *O. latipes* about 145 h after fertilization; S38 = stage 38 in the embryogenesis of *O. latipes* about 200 h after fertilization.

Table 4

Growth of F₁ *Oryzias latipes* exposed to different carbonate alkalinities (CA) after 120 d post-hatch. Mean $\pm$ SE of total length (TL), body weight (BW), specific growth rate of total length (SGR_L); specific growth rate of body weight (SGR_W)

CA (meq l ⁻¹) ¹	n	Mortality (%)	70 d post-hatch		70 d post-hatch – 120 d post-hatch	
			TL (mm)	BW (mg)	SGR _L (% d ⁻¹)	SGR _W (% d ⁻¹)
1.5	47	9.8 $\pm$ 1.3 ^c	1800.00 $\pm$ 400.00	77.10 $\pm$ 20.00	0.61 $\pm$ 0.20	0.74 $\pm$ 0.24
5.3	38	29.0 $\pm$ 2.4 ^b	1625.00 $\pm$ 132.73	43.55 $\pm$ 8.41	0.37 $\pm$ 0.08	0.69 $\pm$ 0.07
8.8	16	46.5 $\pm$ 3.6 ^b	1662.50 $\pm$ 87.50	42.90 $\pm$ 8.00	0.35 $\pm$ 0.12	0.64 $\pm$ 0.07
16.5	–	95.8 $\pm$ 4.2 ^a	–	–	–	–
31.4	–	100.0 $\pm$ 0.0 ^a	–	–	–	–

¹Mean measured concentrations.

n = number of surviving juveniles at 120 d post-hatch.

Different superscript letters in each column = significant differences ($P < 0.05$) after multiple comparisons with Student-Newman-Keuls method.

8.8 meq l⁻¹ for 120 d post-hatch were also significantly higher than in the controls (1.5 meq l⁻¹) ($P < 0.05$). Growth of CA-exposed juveniles at 70 and 120 d were not significantly different in the CA survival treatments. At 70 d post-hatch, no significant differences in body weight and total length were found among the 5.3, 8.8 meq l⁻¹ treatments and the controls. Rates of length gain (SGR_L) and weight gain (SGR_W) from 70 d post-hatch to 120 d post-hatch in the 5.3 and 8.8 meq l⁻¹ treatments were lower than in the controls, but not significantly different ($P > 0.05$) (Table 4).

Reproductive stage

The number of eggs the paired breeder spawned as well as the fertilization rate and egg-hatching rate were significantly lower in the 31.4 meq l⁻¹ treatment than in other treatments (Fig. 2). The numbers of eggs hatched daily per pair in 5.3, 8.8, 16.5 and 31.4 meq l⁻¹ were significantly lower than in the control ($P < 0.05$); only 7 eggs hatched daily per pair in the 31.4 meq l⁻¹ treatment. The breeders in the 1.5–16.5 meq l⁻¹ treatments spawned almost every day, and the mean fertilization rates were $> 50\%$. In the 31.4 meq l⁻¹ treatment, however, the fishes spawned only in the first 14 days within the 60-day test periods, and no fertilized eggs were found in the pairs. Most dead fishes were males exposed to the high concentration treatment (31.4 meq l⁻¹). The highest hatching rate was in the 8.8 meq l⁻¹ treatment. The GSI of female medaka at the end of the reproductive stage was reduced in the 31.4 meq l⁻¹ treatment at 5.8% (GSI), significantly lower ($P < 0.05$) than in other treatments (Fig. 2D). The GSI of males were not calculated because of their high mortality in some treatments.

Discussion

The present study clearly demonstrated the stress effects of CA on the embryos and reproductive performance of *O. latipes*.

CA stress on embryonic development

Numerous ecological and chemical stresses may lead to the deformation of embryos and to high embryonic mortality as well as hatching delays in various species such as common carp (*Cyprinus carpio*) (Stouthart et al., 1995), *Rivulus marmoratus* (Kim et al., 2003), Japanese medaka (Farwell et al., 2006), *Sciaenopus ocellatus* (Gwo et al., 1995), and crayfish (Morris et al., 1989). Stress tolerance of embryos varies at develop-

mental stages and with fish species. Fish embryos at some stages were more sensitive than in others when eggs were subjected to stresses such as mechanical shock and temperature (Gwo et al., 1995). The variation in stress tolerance is probably related to differences in metabolic activity of different cell types among fish species. In the present study, embryo mortality was significantly higher when exposed to high CA (16.5–64.5 meq l⁻¹) than in the control. Although more than 50% of surviving embryos hatched in the 16.5 and 31.4 meq l⁻¹ treatments, most were unable to swim up after hatching. Medaka embryos at the organogenesis stages, from the premature swim bladder stage (S29) to the distinguished cornea (S33) and hatching (S38) stages, were sensitive to CA stress, which is different from other stresses such as mechanical shock and temperature. The egg membrane could provide a shield to prevent intrusion of chemicals. The teleost embryos are protected by two outer protein membranes with a thickness of 10 μ m, the chorion and the inner perivitelline membrane. At the beginning of CA treatment, the chorion prevents the CA from harming the embryo. With continuing exposure, the hatching stage (S38) was also sensitive to CA, because the inner layer of the chorion was dissolved in the hatching stage. Without the chorion, the embryos would not be fully protected when exposed directly to any kind of stressor.

The hearts of pre-hatch larvae appeared to be abnormal in the hatching stage (S38). Sodium bicarbonate was used as an anesthetic on trout and carp (Bookea et al., 1978). This implies that CA stress might decrease the blood flow rate and affect the heart function of medaka.

CA stress on medaka breeder reproduction

Our study found that the growth of CA-exposed juveniles at 70 and 120 d was not affected in the CA survival treatments (5.3 and 8.8 meq l⁻¹), although the reproduction of the medaka breeders was affected (Fig. 2). Many environmental factors such as pH, temperature, photoperiod, food availability and water quality affect the reproduction performance of fish (Wiebe, 1968; Milton and Koehn, 1976; Sawara and Egami, 1977). For example, low pH decreases fecundity and egg fertility of fathead minnows, *Pimphales promelas* (Mount, 1973) and brook trout, *Salvelinus fontinalis* (Menedez, 1976). In some euryhaline fish species, reproductive processes at several stages, such as ovarian development and ovulation, are inhibited in fresh water (Abraham and Blanc, 1966; Eckstein, 1975). Stress also degrades the gamete quality of rainbow trout (Campbell et al., 1992). The development of medaka ovaries at

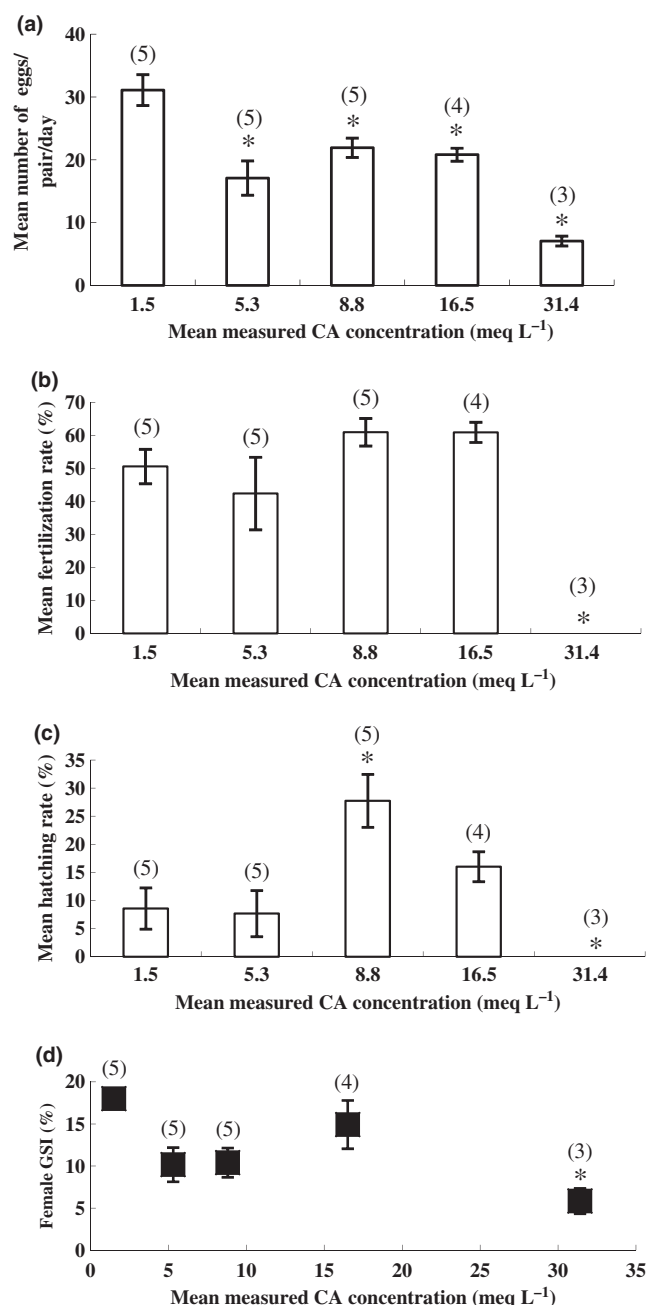


Fig. 2. Number of eggs spawned (a), fertilization rate (b), hatching rate (c) and gonadosomatic index (GSI) (d) of female *Oryzias latipes* for 60-day exposure to different concentrations of carbonate alkalinity (CA). Data expressed as mean $\pm$ SE. Number of breeder pairs in each treatment indicated beyond each bar. *denotes significant differences ($P < 0.05$) after comparison between treatments and controls

the end of the reproductive stage was significantly inhibited in the 31.4 meq L⁻¹ treatment.

Possibilities of using medaka as a model to study alkalinity stress

Our study found that the medaka can adapt to high alkalinities, although these conditions may be stressful to the fish. Some physiological changes regulated by hormone and endocrine systems during alkalinity adaptation should respond to this stress. Techniques in physiological studies such as the removal of the pituitary (Amiya et al., 2007), kidney (Dasmahapatra et al., 2005) and gonad (Yokota et al., 2005) have been achieved, making it possible to use these tissues for

physiological studies. Medaka is also a popular species for genomic research. The relatively small size of the genome (approximately 800 Mb) makes it suitable for genomic analyses (Ishikawa, 2000). With the sequence of medaka draft genome (Kasahara and Naruse, 2007) the number of ESTs in July 2008 exceeds 300 000 sequences, which make genome-wide analysis, such as gene-expressed microarray, possible. All of these make medaka a model for the further study of alkalinity stress.

Summation

Effects of CA stress on medaka embryos and reproduction can be summarized as:

1. Medaka embryos are sensitive to CA stress, especially in stages S29–S33 and S38.
2. Low concentrations (5.3 and 8.8 meq L⁻¹) of the CA treatment did not affect the growth of F₁ medaka.
3. Medaka breeder reproduction was totally inhibited with high CA stress (31.4 meq L⁻¹).
4. Further studies should focus on the gene expression level (transcriptional level) at the embryonic and reproduction stages to reveal the response mechanisms of the medaka to CA stress.

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